

P3 - Paneling, Arrays and Attractor Façade

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Approach: Parametric

Revision: 2026-09-23. Instructor review draft. Grading allocation approved; dates remain proposed for review before publication.

3.1: module, authored material, ground plane and three-point lighting. 3.2: arrays with library or student-authored materials. 3.3: attractor-driven panel façade, shown in elevation and isometric.

3.1 - Descriptive geometry module - material and three-point lighting

Introduced: Week 4 - Sep 15 / **Due:** Week 9 - Oct 20, 11:59 PM Eastern Time / **Weight:** 10%

Description

Construct a panel from a built precedent using explicit point positions and ordered faces. Present three parameter variations on one 17 x 11 inch sheet, using one material and one consistent three-point lighting setup on a ground plane. Include the point list, face list and parameter table. This authored module is reused on Sheets 1-2 of 3.2 and becomes the parameter-driven element in 3.3.

Rules & Constraints

- Choose a simple panel from a built precedent that tiles edge to edge in a 2 x 2 grid. Credit the building, architect and source.
- Define parameters first and explain each. Include a shape control such as aperture, fold depth or corner offset, beyond overall dimensions.
- Build an editable, closed mesh from explicit point and face lists. Calculate point positions from the parameters.
- For a panel with 24 or more points, a Boolean operator is allowed if at least 16 explicit points are used to generate geometry. Review this approach with your professor.
- State the ranges where the module closes and tiles. Diagram low, middle and high values of a shape parameter.
- Use one 17 x 11 inch sheet showing three rendered panel variations with different parameter values. Apply one student-authored module material and one consistent lighting strategy across all three. Use three-point lighting (key, fill and rim/back), a neutral ground plane and no entourage.
- Beside the renders, show the point list, face list referencing those points and a parameter table with the values used for each variation. No separate face-winding explanation or lighting diagram is required.
- Study the [three-point lighting tutorial](#), also listed in Resources. Three-point lighting describes lighting roles; it does not require Point-type lights.

Parameters

- Width, depth and height/thickness in stated model units, with a local origin and X/Y/Z axes.
- At least one shape control beyond overall size: aperture, fold depth, corner offset or an equivalent geometric feature. Identify the control and list the values used for the three variations.
- Valid ranges and dependencies for every control. State the limits within which faces remain valid, the solid closes and neighboring panels meet.

Required Studies

- **Precedent and construction lists.** Credit the building, architect and image source. Show the point list with indices and coordinates, and the face list referencing those point indices. For the permitted Boolean route, identify the geometry generated from at least 16 explicit points and review it with your professor.

- **Three parameter settings.** Render low, middle and high values of a shape parameter using the same graph. Include a parameter table with the values for all three variations. Check closure and edge-to-edge 2 x 2 tiling in the working model; additional construction or tiling diagrams are not required on the sheet.
- **One material and lighting strategy.** Use the same student-authored module material and lighting setup in all three renders. The key light establishes form, the fill controls shadow contrast and the rim/back light separates edges from the background. Keep the module on a neutral ground plane. Follow the [three-point lighting tutorial](#). No lighting diagram is required.

Workflow

1. Select a built panel with a repeatable boundary and identify its points, edges and faces.
2. Define inputs, calculate vertices and assemble ordered face lists. Review any permitted Boolean use with your professor under the 24-point/16-explicit-point rule above.
3. Test closure, normals and tiling at low/middle/high settings. Resolve gaps, reversed faces and intersections.
4. Create one module material and balance the key/fill/rim lights on a ground-plane scene. Render all three variations consistently; compose the sheet with the point list, face list and parameter table.

Deliverables

- **One 17 x 11 inch landscape module sheet:** Lastname_Firstname_P3a.pdf. Include the credited precedent, three rendered panel variations, a point list, a face list referencing those points and a parameter table for the three settings. Use one module material and one lighting strategy throughout.
- **Editable graph/source:** .blend, or an already approved .gh/.3dm route. Include the material/light scene used for the render and package referenced assets. Keep native node names visible and explain your groups with Frames.
- Submit files in Lastname_Firstname_P3a/. Save the chosen parameter setting and panel origin so the exact same module can be reused in P3b. No physical print is required for this milestone.

Evaluation

- **Computational understanding - 30%:** explain how the parameters, point list and face list construct the panel and how a shape input changes it.
- **Visual communication - 30%:** the three renders, material, lighting, lists and parameter table reveal the geometry clearly and follow the Graphic Standards Manual.
- **Technical execution - 25%:** the panel is valid, tiles in the working model and meets the explicit-point/Boolean rules; the editable scene reproduces the submitted views.
- **Exploration - 15%:** three meaningful parameter settings test the shape control and support the selected module for later assignments.

Before You Submit

- [] Built precedent credited; panel authored from calculated points and ordered faces.
- [] One shape control beyond dimensions, tested at low/middle/high values.
- [] Closure, normals and edge-to-edge 2 x 2 tiling checked in the working model.
- [] One sheet shows three rendered variations, the point list, face list and parameter table.
- [] Any Boolean use is reviewed with the professor, for a panel with 24 or more points and at least 16 explicit points used to generate geometry.
- [] One student-authored module material and one key/fill/rim lighting strategy across all variations; neutral ground plane; no entourage. No lighting diagram or material-settings table is required.

3.2 - Arrays

Introduced: Week 6 - Sep 29 / **Due:** Week 10 - Oct 27, 11:59 PM Eastern Time / **Weight:** 10%

Description

Develop basic panel arrays, four conditional façade patterns and an exploratory cube volume shaped by math. Present the three sheets using your Graphic Standards Manual.

Rules & Constraints

- Submit exactly three 17 x 11 inch landscape sheets together as one assignment.
- Sheet 1 - four diagrams: your 3.1 module alone, your module in a 1D array, your module in a 2D array, and your module in a 3D array. Use the same module and label counts, spacing and axes.
- Sheet 2 - four façade-pattern diagrams: use your module in a 2D array and if/Switch conditions to generate four different patterns. Each diagram must identify its condition and resulting modification.
- Sheet 3 - volumetric exploration: use cubes in a 3D array and present an isometric view. Drive vertical variation with a mathematical function of both X and Y, such as sine or cosine. Explore the resulting spatial form; optionally combine it with Switch conditions to generate voids.
- Reuse the 3.1 panel definition on Sheets 1-2. Sheet 2 may transform or omit selected instances to make a pattern. Use cubes, rather than the panel, for the volumetric exercise on Sheet 3.
- Present simple diagrams using one consistent view and material for comparable studies. Keep lighting consistent; separate polished render setups for the four façade patterns are not required. Choose an appropriate view for each dimensional array and use isometric for the cube volume.
- Build repetition from basic nodes and begin with a count you can inspect; there is no minimum element count.
- Use materials from the instructor-provided library or create your own. Record which materials you used and any edits; package required textures. Material choice must keep array spacing, depth and repetition readable. The library will be posted before this assignment is due; creating your own remains an equally valid route.

Parameters

- Integer counts Nx, Ny, Nz; positive spacing dx, dy, dz measured between module origins; base point and array axes. Use one module orientation for Sheet 1.
- Each façade condition, its true/false outcomes and relevant controls, such as row interval, rotation, displacement or keep/remove choice.
- Cube size, layer count, height function and its final settings, including amplitude, frequency and phase where used. Identify any optional void rule.
- Material source and any edits. Library or student-authored materials are permitted; package textures.

Required Studies

- **Sheet 1 - Module and dimensional repetition.** Show the module, row, grid and spatial array. Label counts, spacing, axes and totals Nx, Nx*Ny and Nx*Ny*Nz. Make the 3D layers visible within that diagram.
- **Sheet 2 - Four conditional façade patterns.** Explore four distinct patterns using if/Switch. Possible rules include alternating rows or columns, checkerboards and diagonal bands. Rotate, move, scale or omit selected instances. Label each condition and its true/false outcomes.
- **Sheet 3 - Math shapes a cube volume.** Explore a 3D array of cubes, using the Sine curve scene in the Arrays Part 2 example file as a reference. One possible relationship is $f(x, y) = A \sin(fx \cdot x + px) + B \cos(fy \cdot y + py)$ with layer positions $z = k \cdot dz + f(x, y)$. Here k is the layer index; angles are in radians. Retain multiple layers. Explore the settings and select one result for the isometric. Switch-based voids are optional.

Workflow

1. Reference your 3.1 module. Build a row, then nested 2D and 3D arrays from basic nodes; check their counts.
2. Develop four façade patterns from the 2D array. Verify that each condition affects the intended instances.

3. Build a separate layered cube array. Explore functions of X and Y, optionally add voids, and save the settings of your selected result.
4. Check for unintended collisions and duplicate positions. Compose the three sheets with readable views, rule captions and materials.

Deliverables

- **Exactly three 17 x 11 inch landscape sheets in one PDF:** Lastname_Firstname_P3b.pdf. They are one assignment, submitted together.
- **Sheet 1 - Four diagrams.** Your module alone, then the same module in a 1D, 2D and 3D array. Label axes, counts, spacing and array totals.
- **Sheet 2 - Four façade patterns.** Four diagrams using your module in a 2D array, modified by if/Switch conditions. Include a short condition and outcome caption for each pattern.
- **Sheet 3 - Cube volume.** One isometric view of an exploratory 3D cube array with vertical variation driven by a function of X and Y. Include the formula and final settings; describe any optional rule used to create voids.
- **Editable graph/source** in Lastname_Firstname_P3b/, with labeled inputs, saved settings and outputs for all three sheets. Package dependencies, textures and a material-source note. No physical print is required.

Evaluation

- **Computational understanding - 30%:** explain dimensional repetition, if/Switch conditions and the function of X and Y; demonstrate how their controls change the outputs.
- **Visual communication - 30%:** the three sheets reveal repetition, façade patterns and the cube volume; captions and diagrams follow the Graphic Standards Manual.
- **Technical execution - 25%:** array totals agree, conditions affect the intended instances and saved inputs reproduce every output; Sheets 1-2 use the panel and Sheet 3 uses layered cubes.
- **Exploration - 15%:** four distinct façade patterns and purposeful function tests support the selected spatial result. Conditional voids remain optional.

Before You Submit

- [] Sheet 1 has exactly four diagrams showing the 3.1 module alone and in 1D, 2D and 3D arrays.
- [] Axes, counts, spacing and totals are labeled for the basic arrays.
- [] Sheet 2 has four different façade patterns using the module, each with an if/Switch rule and outcomes.
- [] Sheet 3 shows an isometric cube volume, with multiple layers and vertical variation driven by X and Y.
- [] The height function and selected values are recorded; any optional void rule is identified.
- [] Three sheets submitted together; the source reopens and reproduces all documented outputs.
- [] Material library selections or student-authored materials identified; referenced textures packaged.

3.3 - Attractor-driven panel façade

Introduced: Week 9 - Oct 20 / **Due:** Week 13 - Nov 17, 11:59 PM Eastern Time / **Weight:** 10%

Preparation checks: Week 10 - Oct 27: Bring a uniform façade and a first test combining one curve and one or two point attractors to drive one panel parameter.

Description

Design a responsive façade using one curve attractor and one or two point attractors together. Combine their influences to change one geometric parameter in your 3.1 panel. Show the overall composition in elevation and its three-dimensional result in isometric.

Rules & Constraints

- Use your 3.1 panel and the 2D array logic from 3.2 to build a façade. Use one curve attractor together with one or two point attractors. At each panel location, measure the shortest distance to the curve and the distance to each point.
- Use the attractor data to drive exactly one geometric parameter of the panel, such as aperture, fold depth or corner offset. Recompute panel geometry from that parameter. Color alone or moving unchanged panels does not satisfy this requirement.
- Remap the curve and point distances to influence values and combine them using a stated rule, such as an average or weighted blend. Both the curve and the point attractor(s) must contribute. Diagram the combination and clamping into the valid parameter range tested in 3.1; keep other panel inputs fixed.
- Compare a uniform baseline with the combined-attractor façade. Show one additional configuration by moving one point or editing the curve, while keeping the remaining attractors and mapping settings fixed.
- Present one 17 x 11 inch composition with an orthographic façade elevation and a matching isometric view of the same final façade. Show the curve, every point attractor and the combined influence in an overlay or accompanying field diagram with a legend.
- Submit a print-ready PDF and the editable graph. These are views of a 3D panel system; a standalone 2D line-field drawing is no longer the assignment. Pen plotting is not required.

Parameters

- Façade row/column counts, panel spacing and base plane/origin. Use a stable grid so the comparison isolates changes to panel form.
- One attractor curve and one or two attractor point positions. Record distance ranges, falloff, combination rule and any weights; clamp the combined influence before mapping to panel geometry.
- One selected geometric panel parameter, its units and minimum/maximum values. Keep the other shape parameters fixed; test validity at both extremes.

Required Studies

- **Baseline.** Show a small uniform façade with the selected panel parameter fixed. Label the constant value.
- **Field to form.** Diagram curve/point distances to remapped influences to combination to clamped panel parameter. Show the combined field, curve and point locations with a clear legend.
- **Controlled variation.** Move one point or edit the curve once. Keep the grid, other attractors, mapping and remaining panel inputs fixed so the effect is clear.
- **Final composition.** Give the elevation and matching isometric the most space. Identify the curve, all point attractors, driven parameter and final settings. Use the Graphic Standards Manual.

Workflow

1. Reference your own panel definition and create a small façade grid. Confirm the uniform baseline first.
2. Measure each panel's distance to the curve and point(s). Remap and combine their influences, clamp into the chosen parameter range, and generate the panels. Check that each attractor affects the result.

3. Check the near and far limits and several intermediate panels. Resolve gaps, self-intersections or invalid faces before increasing the array count.
4. Capture the two combined-attractor configurations and compose the elevation, isometric and diagrams from the saved final state.

Deliverables

- **One 17 x 11 inch landscape PDF:** Lastname_Firstname_P3c.pdf. Include the final façade elevation and isometric, the curve and point locations, combined-field legend, mapping diagram, baseline and one changed-attractor configuration.
- **Editable source** in Lastname_Firstname_P3c/: save the panel definition, array, curve, point(s), combination and mapping controls; package all dependencies and materials. Record the settings used for the final views.
- Prepare the composition for normal printing at its stated size. No pen-plotter output or physical façade model is required.

Evaluation

- **Computational understanding - 30%:** explain how the curve and point influences combine to drive one panel parameter; demonstrate the effect of moving one point or editing the curve.
- **Visual communication - 30%:** elevation, isometric, field legend and captions explain the façade composition and follow the Graphic Standards Manual.
- **Technical execution - 25%:** one curve and one or two point attractors contribute to a valid panel response; the packaged source and saved settings reproduce the final views.
- **Exploration - 15%:** baseline and changed-attractor studies isolate the control's effect and support the selected composition.

Before You Submit

- [] Own panel and array reused; one curve and one or two point attractors jointly drive one panel parameter.
- [] Curve and point locations, ranges, combination rule, units and clamping are documented.
- [] Uniform baseline and one additional attractor setting are shown.
- [] Final elevation and isometric show the same façade; field influence is legible.
- [] Print-ready composition and packaged source reproduce the final result.

Shared standards: Course software, fabrication, attribution and AI policies apply. Submit through the channel announced in class/Canvas. Test the packaged source from a clean folder. Percentages follow the instructor review draft. This handout uses US Letter; student sheets are 17 x 11 inches.

Midterm grade: The midterm grade accumulates the Graphic Standards Manual, CSG sheet and print, module (3.1) and arrays (3.2). These assignments account for 40% of the final course grade. Report the accumulated result as a percentage of those 40 possible course points. The midterm adds no separate assessment weight.